# 💻 Computer Networking

## 📚 Definition

Computer networking is the process of **connecting two or more computers** to **share data, resources, and services** (like files, printers, or internet).

## 📡 Main Purpose

- Communication
- Resource sharing
- Data sharing
- Remote access

## 🔗 Types of Networks

1. **LAN (Local Area Network):** Small area (home, school).
2. **MAN (Metropolitan Area Network):** Covers a city.
3. **WAN (Wide Area Network):** Covers large areas (like the Internet).

## 💻 Network Devices

- **Router:** Connects different networks.
- **Switch:** Connects devices in a LAN.
- **Hub:** Sends data to all devices.
- **Modem:** Connects to the internet.

## 🧠 Examples

- School computer lab network
- Internet
- Office network

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# 🌐 IPv4 Address

## 📚 Definition

IPv4 (Internet Protocol version 4) is the **4th version** of the Internet Protocol used to **identify devices on a network** using a **unique 32-bit address**.

## 🔢 Format

- Written as **four numbers** separated by dots (e.g., `192.168.1.1`)
- Each number = **8 bits (1 byte)**
- Total = **32 bits = 4 bytes**

### 📡 Address Range

`0.0.0.0` → `255.255.255.255`

### 🔗 Types of IPv4 Addresses

1. **Public IP:** Used on the internet.
2. **Private IP:** Used inside local networks.
3. **Static IP:** Fixed address.
4. **Dynamic IP:** Changes automatically.

### 🏷️ Classes of IPv4

| Class | Range | Use |
|---|---|---|
| A | 1.0.0.0 – 126.255.255.255 | Large networks |
| B | 128.0.0.0 – 191.255.255.255 | Medium networks |
| C | 192.0.0.0 – 223.255.255.255 | Small networks |
| D | 224.0.0.0 – 239.255.255.255 | Multicasting |
| E | 240.0.0.0 – 255.255.255.255 | Experimental |

### 💡 Example

`192.168.0.1` → Private IP (Class C)

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# 🧠 Subnet Mask

## Definition

A **subnet mask** is a **32-bit number** that divides an IP address into two parts:

- **Network part** – identifies the network.
- **Host part** – identifies the specific device in that network.

### 🧮 Example

```
IP Address:   192.168.1.10
Subnet Mask:  255.255.255.0
Binary:       11111111.11111111.11111111.00000000
```

☑️ The first 24 bits (1s) → Network
☑️ The last 8 bits (0s) → Host

### 📊 Purpose

- To separate **networks** and **hosts**
- To use IP addresses efficiently
- To improve **network management** and **security**

## 🧩 Common Subnet Masks

| CIDR | Subnet Mask | No. of Hosts |
|---|---|---|
| /8 | 255.0.0.0 | 16,777,214 |
| /16 | 255.255.0.0 | 65,534 |
| /24 | 255.255.255.0 | 254 |

## ⚙️ Formula to Find Hosts

```
Hosts = 2^(32 - Network bits) - 2
```

**Example:**
For `/24` → `2^(32 - 24) - 2 = 254 hosts`

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# 🌍 Domain Name System (DNS)

## 📓 Definition

DNS (Domain Name System) is a **naming system** that **converts domain names** (like `www.google.com`) into **IP addresses** that computers use to identify each other on the internet.

## 🌍 Purpose

- Makes it easier to access websites without remembering IPs.
- Acts like the **"phonebook" of the internet**.

## 🔗 How It Works

1. User types a website name.
2. DNS server translates the name into an IP address.
3. Browser connects to that IP to load the website.

## 🧠 Important Terms

- **Domain Name:** Human-readable name (e.g., `google.com`)
- **DNS Server:** Translates domain to IP
- **IP Address:** Numerical address of a website
- **Resolver:** Finds the IP address requested by the user

## 🏷️ Hierarchy of DNS

1. **Root Domain (.)**
2. **Top-Level Domain (TLD):** `.com`, `.org`, `.in`
3. **Second-Level Domain:** `google` in `google.com`
4. **Subdomain:** `mail.google.com`

### 💡 Example

`www.yahoo.com` → DNS → `98.137.11.163`

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## 🧠 How DNS Servers Work

### 📘 Definition

DNS servers are systems that **translate domain names into IP addresses** so browsers can locate and connect to websites.

### ⚙️ Working Steps (Process)

1. **User Request:**
   You type `www.google.com` in the browser.
2. **DNS Resolver:**
   The request goes to a **DNS resolver** (usually your ISP's server).
3. **Root Server:**
   Resolver asks the **Root DNS Server** where to find the Top-Level Domain (TLD) server (`.com`).
4. **TLD Server:**
   TLD server gives the address of the **Authoritative DNS Server** for `google.com`.
5. **Authoritative Server:**
   It provides the **actual IP address** of `www.google.com`.
6. **Browser Connects:**
   The resolver sends that IP back to your browser, which connects to the website.

### 💡 Example

`www.google.com` → DNS Resolver → Root Server → TLD Server → Authoritative Server → IP Address → Website loads

### 🧠 Key Servers in DNS Process

- **Resolver:** Finds IP on behalf of the user.
- **Root Server:** Directs to correct TLD server.
- **TLD Server:** Points to the domain's authoritative server.
- **Authoritative Server:** Holds the final IP information.

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## 🔢 Port Numbers

## 🧾 Definition

A **port number** is a **numerical identifier** that helps a computer **distinguish between different types of network services or processes** running on the same device.

## 💡 Purpose

- Allows multiple network applications to use the same IP address.
- Identifies **specific services** (like web, email, FTP).

## 🔢 Range of Port Numbers

- **0 – 65535** (Total 65,536 ports)

## 🧩 Types of Ports

1. **Well-known Ports (0–1023):** Used by common services
    - HTTP → 80
    - HTTPS → 443
    - FTP → 21
    - SSH → 22
    - DNS → 53
2. **Registered Ports (1024–49151):** Used by user or custom apps
3. **Dynamic / Private Ports (49152–65535):** Temporary ports for client communication

## 🌐 Example

`https://www.example.com:443` → Port 443 used for HTTPS

## 📦 Note

Each application on a device uses a **unique port number** to send and receive data.

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# 🌐 Network Interfaces

## 🧾 Definition

A **network interface** is the **point of connection** between a computer and a network. It allows the device to **send and receive data** over that network.

## 💡 Purpose

- Connects a device to LAN, WAN, or the internet.
- Handles communication between hardware and network software.

## 🧩 Types of Network Interfaces

1. **Wired Interface:** Uses cables (like Ethernet).

2. **Wireless Interface:** Uses Wi-Fi or Bluetooth.
3. **Virtual Interface:** Created by software (e.g., VPN, virtual machines).

### 💻 Examples

- **Ethernet card (NIC)** – for wired connection
- **Wi-Fi adapter** – for wireless connection
- **Loopback interface (127.0.0.1)** – for internal testing

### ⚙️ NIC (Network Interface Card)

- Hardware device that provides network connectivity.
- Has a unique **MAC address** (Media Access Control).

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## 🏷️ MAC Address

### 🧾 Definition

A **MAC address** (Media Access Control address) is a **unique hardware identifier** assigned to a device's **network interface card (NIC)** for communication within a network.

### 🔢 Format

- 48-bit (6 bytes) address.
- Written in **hexadecimal**, divided by colons or hyphens.
  👉 Example: `00:1A:2B:3C:4D:5E`

### 💡 Purpose

- Identifies devices uniquely on a local network.
- Used in data link layer (Layer 2) of the OSI model.
- Helps in transferring data packets within the same network.

### 🧩 Types of MAC Address

1. **Unicast:** Identifies a single device.
2. **Multicast:** For a group of devices.
3. **Broadcast:** For all devices in a network.

### ⚙️ Key Points

- Assigned by the device manufacturer.
- Cannot be changed (but can be spoofed).
- Works within **LAN**, not over the internet.

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## 🔥 Firewalls

## 📝 Definition

A **firewall** is a **security system** that **monitors and controls incoming and outgoing network traffic** based on **predefined security rules**.

## 💡 Purpose

- Protects a computer or network from **unauthorized access**.
- Blocks **malicious traffic** and allows **safe communication**.

## 🧩 Types of Firewalls

1. **Packet Filtering Firewall:** Checks each data packet and allows or blocks it.
2. **Proxy Firewall:** Acts as a gateway between users and the internet.
3. **Stateful Inspection Firewall:** Monitors the state of active connections.
4. **Next-Generation Firewall (NGFW):** Advanced; includes antivirus, intrusion prevention, etc.

## 🌐 Works On

- **Network Layer** and **Transport Layer** of the OSI model.

## ⚙️ Functions

- Filters harmful data packets.
- Prevents hacking and malware attacks.
- Monitors all incoming/outgoing traffic.

## 💡 Examples

- Windows Defender Firewall
- Cisco ASA

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# 🌐 OSI Model

## 📝 Definition

The **OSI (Open Systems Interconnection) model** is a **conceptual framework** that defines **how data travels** through a network in **7 layers**.

## 💡 Purpose

- Helps understand and design network communication.
- Standardizes how devices communicate.

## 🔢 7 Layers of OSI Model (Top to Bottom)

1. **Application (Layer 7):** User interface (e.g., HTTP, FTP).
2. **Presentation (Layer 6):** Data translation, encryption, compression.

3. **Session (Layer 5):** Manages sessions and connections.
4. **Transport (Layer 4):** Ensures reliable data transfer (TCP/UDP).
5. **Network (Layer 3):** Handles IP addressing and routing.
6. **Data Link (Layer 2):** Error detection, MAC addressing.
7. **Physical (Layer 1):** Transmission of raw bits (cables, signals).

### 🧠 Mnemonic

👉 *All People Seem To Need Data Processing* (Top to Bottom)

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# ✅ TCP/IP Model

### 📚 Definition:

The **TCP/IP model (Transmission Control Protocol / Internet Protocol)** is a networking framework used to connect devices on the internet.
It defines how data is transmitted and received.

### 💡 Purpose:

- Basis of the modern Internet.
- Helps computers communicate over networks reliably.

### 🔢 Layers of TCP/IP Model (4 Layers):

1. **Application Layer** – User interaction (HTTP, FTP, DNS, SMTP).
2. **Transport Layer** – Reliable delivery (TCP, UDP).
3. **Internet Layer** – Handles addressing and routing (IP).
4. **Network Access Layer** – Physical transmission (Ethernet, Wi-Fi).

### 🧠 Mnemonic:

👉 **A Tiny Intelligent Network**
*(Application → Transport → Internet → Network Access)*

### 📦 Comparison with OSI Model:

| TCP/IP Layer | OSI Layers |
| --- | --- |
| Application | Application, Presentation, Session |
| Transport | Transport |
| Internet | Network |
| Network Access | Data Link, Physical |

### 💡 Example Protocols:

- **Application Layer:** HTTP, FTP, DNS
- **Transport Layer:** TCP, UDP
- **Internet Layer:** IP, ICMP
- **Network Access Layer:** Ethernet, Wi-Fi

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# ☑ Network Topologies

## 📘 Definition:

Network topology refers to the layout or arrangement of devices (nodes) and how they are connected in a network.

## 💡 Types:

1. **Bus** – One main cable; cheap but hard to troubleshoot.
2. **Star** – All connect to a hub; reliable.
3. **Ring** – Circular connection; one failure affects all.
4. **Mesh** – Every device connected to all; costly but strong.
5. **Tree** – Hierarchical mix of star + bus.
6. **Hybrid** – Combo of topologies; flexible & scalable.

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# ☑ TCP and UDP

## 📘 Definition:

**TCP (Transmission Control Protocol)** and **UDP (User Datagram Protocol)** are **Transport Layer protocols** used for data transmission over the Internet.

## 💡 TCP (Transmission Control Protocol):

- **Connection-oriented** – establishes a connection before data transfer.
- **Reliable** – ensures data is delivered in order.
- **Error checking & flow control** available.
- **Slower** but **accurate**.
- 3-way handshake (connection setup)
- Acknowledgments (ACKs)
- 📦 *Used in:* Web (HTTP/HTTPS), Email (SMTP), File transfer (FTP).

## 💨 UDP (User Datagram Protocol):

- **Connectionless** – no setup before sending data.
- **Unreliable** – data may be lost or out of order.
- **No error checking or acknowledgment.**
- **Faster** but less reliable.
- 🎮 *Used in:* Video streaming, online games, voice calls (VoIP).

### 🧠 Key Difference:

| Feature | TCP | UDP |
|---|---|---|
| Connection | Connection-oriented | Connectionless |
| Reliability | Reliable | Unreliable |
| Speed | Slower | Faster |
| Error Checking | Yes | No |
| Example | HTTP, FTP | DNS, YouTube, Games |

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# 💻 Accessing Remote Terminal Using SSH

### ◇ What is SSH?

- **SSH (Secure Shell)** is a protocol used to securely connect to remote systems.
- It allows you to **access, control, and manage** another computer over the internet or a network.

### ◇ How it Works

1. Connect using:

```
ssh username@remote_ip
```

2. Authenticate using password or SSH key.
3. Gain remote terminal access to run commands.

### ◇ Benefits

- 🔒 Encrypted and secure
- 🧑‍💻 Remote management
- 🧱 Protects against data interception

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# 🗁 Transferring Files Using SCP

### ◇ What is SCP?

- **SCP (Secure Copy Protocol)** securely copies files between **local and remote systems**.
- It uses SSH for secure data transfer.

### ◇ Basic Commands

- **Upload (local → remote):**

```
scp file.txt username@remote_ip:/path/to/destination
```

- **Download (remote → local):**

```
scp username@remote_ip:/path/to/file.txt /local/path
```

- **Copy folder (recursive):**

```
scp -r folder username@remote_ip:/path/to/destination
```

### ◇ Benefits

- 🔒 Secure with SSH encryption
- ⚡ Fast and easy for file transfers

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# 🔑 SSH Public and Private Keys

### ◇ Key Pair Concept

- SSH uses **asymmetric encryption** with two keys:
  - **Public Key:** Shared with the server
  - **Private Key:** Kept secret on your local system

### ◇ Authentication Process

1. Generate key pair:

```
ssh-keygen
```

2. Copy public key to the server:

```
ssh-copy-id username@remote_ip
```

3. SSH verifies your **private key** to log in without a password.

### ◇ Advantages

- 🚫 No password needed for every login
- 🔒 More secure than password authentication

- 🧩 Protects against brute-force attacks

# 🧠 Summary

| Concept | Purpose | Command Example |
|---|---|---|
| **SSH** | Remote terminal access | `ssh user@ip` |
| **SCP** | Secure file transfer | `scp file user@ip:/path` |
| **SSH Keys** | Passwordless secure login | `ssh-keygen`, `ssh-copy-id` |